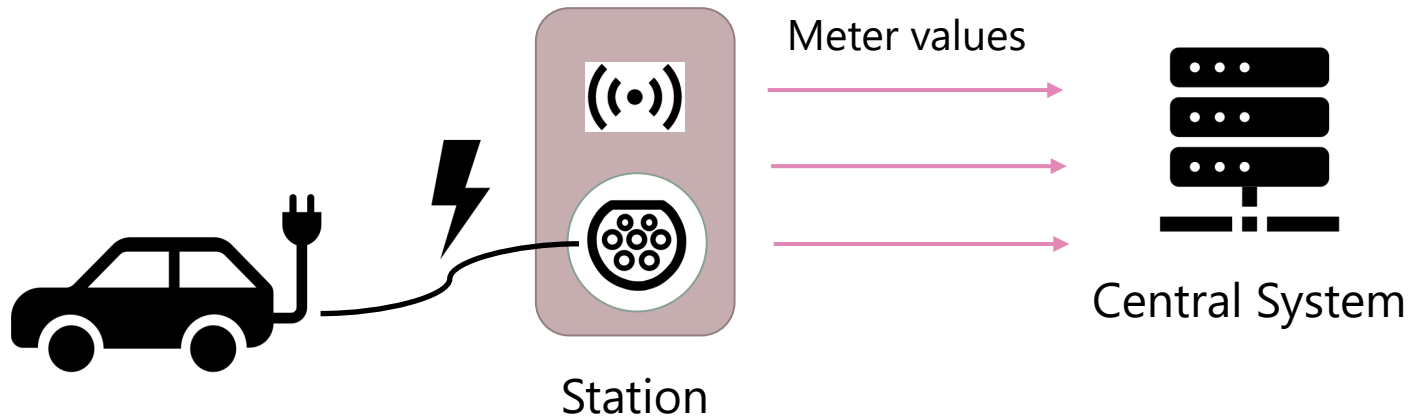


7. METER VALUES



Information about the ongoing charge



While charging is in progress, the station keeps the central system updated by sending **meter values**. The most common meter value is the amount of energy (in **watt-hours**) that has been charged so far.

Sending meter values is **optional** in the OCPP standard, but in practice, all stations send at least **energy readings** during charging. This data is typically used, for example, to show the amount of energy charged in mobile apps.

It is also possible to specify the **interval** at which different meter values are sent. Most stations use the "interval" parameter in the "BootNotification" response as the default meter value interval.



Basic structure of a Metervalues.req message

Example:

```
[2, "533-aa-22919" , "MeterValues",  
  { "connectorId":1,  
    "transactionId": 1191,  
    "meterValue": [  
      { "timestamp": "2022-12-08T00:20:40Z",  
        "sampledValue": [  
          { "value": "2901.000",  
            "context": "Sample.Periodic",  
            "measurand": "Energy.Active.Import.Register",  
            "unit": "Wh" } ]  
        }  
      ]  
    }  
  ]  
]
```

Structure of **Metervalues.req** message is a bit more complex than many other OCPP messages. The first two fields, "**connectorId**" and "**transactionId**," indicate the connection and transaction to which the meter values are related. The next field, "**meterValue**," is an array that contains one or more different meter values

- Each meter value has a "**timestamp**" field that indicates when the measurement was taken.
- A station can measure several "**sampledValues**" at the same time, for example, energy (in watt-hours) and current (in amperes).
- Each "**sampledValue**" has information about the actual **value**, **context**, **measurand**, and **unit**.



More complex example

```
[2, "533-aa-22920", "MeterValues",  
  { "connectorId": 1, "transactionId": 1191,  
    "meterValue": [  
      { "timestamp": "2022-12-08T00:20:40Z", "sampledValue": [  
        { "value": "2901.000", "context": "Sample.Periodic", "measurand": "Energy.Active.Import.Register", "unit": "Wh" },  
        { "value": "14.7", "context": "Sample.Periodic", "measurand": "Current.Import", "unit": "A", "phase": "L1" },  
        { "value": "0", "context": "Sample.Periodic", "measurand": "Current.Import", "unit": "A", "phase": "L2" },  
        { "value": "0", "context": "Sample.Periodic", "measurand": "Current.Import", "unit": "A", "phase": "L3" },  
        { "value": "232.300", "context": "Sample.Periodic", "measurand": "Voltage", "unit": "V", "phase": "L1-N" } ]  
      } ]  
    } ]  
  }
```

In this example, the station reports several different meter values: the charged **energy** in this transaction is **2901.0 watt-hours**, the station has a 3-phase electricity connection (L1-L3), but at the moment it seems that the EV is only using the **L1** phase for charging, as the charging current on L1 is **14.7 amperes** and the other **L2-L3** phases are **zero**. It also states that the **voltage** of phase **L1** is **232.300 volts**.



MeterValues.req – Common Fields

Field	Card.	Description
connectorId	1..1	Required, value ≥ 0 . This is the number of the connector from where the meter values are being reported from. Connector 0 is used to refer to the main power meter.
transactionId	0..1	Optional. Tells the ID of the transaction where the meter values are related to. Even if it is optional by the standard, it is really commonly used by the stations.
meterValue	1..*	Required. Details of the sampled meter values (for example energy, current, voltage etc.)

meterValue – Details of the different sampled values

Field	Card.	Description
timestamp	1..1	Required. Timestamp when the station measured the sampled values
sampledValue	1..*	Required. Details of the actual sampled values that were measured at the time of the timestamp (for example energy, current, voltage etc.)



MeterValues.req -> meterValue -> sampledValue

Field	Card.	Description
value	1..1	Required. The actual value of the measurand. For example how much energy is used, what the voltage is at the moment etc. Example: "42.57"
context	0..1	Optional. Tells in which context the measurement was made. Most typical value is "Sample.Periodic", which tells that the measurement is relative to the start time of transaction. Other typical values could be "Transaction.Begin" or "Transaction.End" which tell that values were taken at the beginning or end of a transaction.
format	0..1	Optional. Default value is "Raw" which tells that the value is in normal raw format, for example as "42.57". Other option is "Signed" which tells that the value field has data in a signed binary format.
measurand	0..1	Optional. Tells what type of value we are measuring. Default value is "Energy.Active.Import.Register" that tells that we are measuring how much energy has been charged. Other typical values could be "Voltage", "Current" or "Temperature"
phase	0..1	Optional. If the measurand is used to measure electrical current phase related things, such as Current, this tells which phase the measurement is related to. Example: "L1"
location	0..1	Optional. Default value is "Outlet", telling that measurement was taken by charging station's outlet. Other value could be for example "Body", telling that for example temperature measurement was taken from inside the charging station's body.
unit	0..1	Optional. Unit of the measurand, for example "Wh", "A", "V", etc. Default is "Wh".



Note on Sampled Values

- Most fields in "**sampledValues**" are **optional**. The simplest way to report energy readings is to only include the "value" field with the energy reading in watt-hours and leave the other fields blank.

```
[2, "533-aa-22920" , "MeterValues",  
  { "connectorId":1, "transactionid": 1191,  
    "metervalue": [  
      { "timestamp": "2022-12-08T00:20:40Z",  
        "sampledValue": [ { "value": "2901.000" } ]  
      }  
    ]  
  }  
]
```

- Many fields in "sampledValues" are **enumerated**, with allowable values defined in the OCPP standard. For a complete list of values for these fields, see Part 7.43 of the OCPP documentation.

